

Sara Baradaran

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EDUCATION

Ph.D. in Computer Science, GPA: 4.0/4.0 University of Southern California	Aug 2023–Present Los Angeles, USA
M.Sc. in Computer Science, GPA: 4.0/4.0 University of Southern California	Aug 2023–Dec 2025 Los Angeles, USA
B.Sc. in Computer Engineering, GPA: 19.49/20 (or 4.0/4.0) Isfahan University of Technology	Sep 2017–Feb 2022 Isfahan, Iran

RESEARCH INTERESTS

Program Analysis	Software Security	Software Testing and Verification
Binary Analysis	Automated Reasoning	ML/AI for Software Engineering

RESEARCH & WORK EXPERIENCE

Graduate Research Assistant University of Southern California, Program Reasoning Lab – Worked on probabilistic Bayesian reasoning for software fault localization and type inference. – Designed SemQL, an extension of the CodeQL query language for writing semantic queries over code.	Aug 2023–Present Los Angeles, USA
Undergraduate Research Assistant Isfahan University of Technology, Intelligent Mobile Networks Lab – Designed and implemented a testbed for intelligent 5G networks.	Nov 2022–Feb 2023 Isfahan, Iran
Undergraduate Research Assistant Isfahan University of Technology, Software Security Lab – Developed an automated framework for detecting software vulnerabilities in binary executables.	Dec 2020–Oct 2022 Isfahan, Iran

PUBLICATIONS

- [1] **S. Baradaran**, Y. Huang, W. Le, and M. Raghothaman, “Prosecutor: Bayesian Counterfactual Fault Localization”, *Accepted at the Proceedings of the ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2026.
- [2] **S. Baradaran**, A. Nazari, and M. Raghothaman, “LLM-Integrated Declarative Program Analysis”, in *Proceedings Of The 15th ACM SIGPLAN International Workshop On The State Of The Art In Program Analysis (SOAP)*, 2026. DOI: 10.1145/3814987.3814993
- [3] **S. Baradaran** and M. Raghothaman, “Effective Minimization of Failure-Inducing Tests Using Convention-Aware Slicing”, in *IEEE International Conference on Software Analysis, Evolution and Reengineering - Companion (SANER-C)*, 2026, pp. 349–356. DOI: 10.1109/SANER-C67878.2026.00053
- [4] **S. Baradaran**, L. Huang, M. Raghothaman, and W. Wang, “Reusing Legacy Code in Wasm: Key Challenges of Compilation and Code Semantics Preservation”, *IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, pp. 181–192, 2026. DOI: 10.1109/SANER67736.2026.00029
- [5] S. Ganji, S. Behnaminia, A. Ahangarpour, E. Mazaheri, **S. Baradaran**, Z. Zali, M. R. Heidarpour, A. Rakhshan, and M. F. Shoyari, “CN2F: A Cloud-Native Cellular Network Framework”, *Cluster Computing*, vol. 28, p. 493, 2025. DOI: 10.1007/s10586-025-05155-w
- [6] **S. Baradaran**, M. Heidari, A. Kamali, and M. Mouzarani, “A Unit-based Symbolic Execution Method for Detecting Memory Corruption Vulnerabilities in Executable Codes”, *International Journal of Information Security*, vol. 22, pp. 1277–1290, 2023. DOI: 10.1007/s10207-023-00691-1
- [7] M. Mouzarani, A. Kamali, **S. Baradaran**, and M. Heidari, “A Unit-based Symbolic Execution Method for Detecting Heap Overflow Vulnerability in Executable Codes”, in *Tests and Proofs*, L. Kovács and K. Meinke, Eds., Cham: Springer International Publishing, 2022, pp. 89–105. DOI: 10.1007/978-3-031-09827-7_6

ACADEMIC SERVICES

Journal Reviewer:

- Transactions on Software Engineering and Methodology (TOSEM): 2025, 2026
- IEEE Access: 2026

Artifact Evaluation Committee

- The IEEE Symposium on Security and Privacy (IEEE S&P): 2027
- The ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL): 2027
- The ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA): 2026
- The ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI): 2026
- The IEEE Annual Computer Security Applications Conference (ACSAC): 2025

TEACHING EXPERIENCE

Teaching Assistant

University of Southern California	Los Angeles, USA
– CSCI 350, Introduction to Operating Systems	Spring 2025
– CSCI 530, Security Systems	Fall 2024

Teaching Assistant

Isfahan University of Technology	Isfahan, Iran
– ECE 34102, Advanced Programming Lab	Spring 2023
– ECE 34304, Operating Systems Lab	Fall 2021, Fall 2022
– ECE 30400, Fundamentals of Computer Security	Spring 2021
– ECE 34333, Compiler Design	Spring 2020, Fall 2020, Spring 2022
– ECE 34425, Algorithm Design and Analysis	Spring 2020, Spring 2022
– ECE 30217, Discrete Mathematical Structures	Spring 2019
– ECE 30151, C Programming Lab	Fall 2018, Spring 2019

HONORS & AWARDS

- **Ranked 1st** in the Electrical and Computer Engineering Department of Isfahan University of Technology, among more than 280 undergraduate students 2017–2022
- **Ranked in the top 0.5%** in Iran’s National Entrance Exam for B.Sc. programs in Mathematics and Physics
- **Placed 2nd** in the Junior Soccer Open Weight League at the RoboCup IranOpen International Competitions Apr 2015

SOFTWARE PROJECTS

See full list of projects on GitHub

- **UbSym.** Framework for detecting memory corruption vulnerabilities in x86-64 binary codes.
- **Wuzzer.** Web fuzzer for detecting XSS, SQL, and CMD injection vulnerabilities in websites.
- **WasmChecker.** Framework for differential testing between C/C++ programs and their WebAssembly counterparts.
- **Prosecutor.** Fault localization tool for Java software which leverages counterfactual reasoning and Bayesian inference.
- **FitSlicer.** A Java code slicer for effective test case minimization.
- **SemQL.** An extension of the CodeQL query language for writing semantic queries over code.

SKILLS

- **Programming Languages:** Python, C, C++, Java, Rust, Solidity, MATLAB, R, JavaScript
- **Database Languages:** SQL, Datalog
- **Assembly Languages:** MIPS, x86/x64, WebAssembly, AVR
- **Frameworks & Libraries:** CodeQL, Angr, Soot, Doop, Beautiful Soup, Scikit, PyTorch, TensorFlow, Keras, Scapy, Selenium, Qt
- **Tools:** LLVM, GDB, Docker, Git, Burp Suite, Wireshark, Nmap, Cisco Packet Tracer, Flex, Bison, Microsoft SQL Server
- **Operating Systems:** Linux, Windows, macOS
- **Technical Skills:** Program Analysis, Agentic Software Engineering, Software & Web Security Analysis, Software Testing & Verification, Machine Learning, Binary Analysis